Cisco PE Node KOR-PE-01 Koramangala Datacenter Hardware/Environmental Fault SOP

Node: KOR-PE-01 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_8612

Description

The Cisco PE node KOR-PE-01 at the Koramangala Datacenter has experienced a hardware/environmental fault resulting in an ASIC memory parity error, line card crash, or power supply redundancy loss, triggering a warning alarm ALARM_8612. This failure domain is a critical concern, as it can compromise the performance, reliability, and overall integrity of the network infrastructure. The PE node's role in this context is to function as a provider edge router, aggregating and forwarding traffic between the provider's network and its customers. The underlying protocol responsible for this failure is the Asynchronous Transfer Mode (ATM) protocol, which relies on the Line Card (LC) to process and forward ATM cells. The Line Card's failure or ASIC memory parity error can lead to a loss of connectivity and subsequently trigger the alarm. The power supply redundancy loss is also a critical failure, as it can cause an immediate shutdown of the node, thereby disrupting network services. In this SOP, we will walk through the steps to diagnose, resolve, and verify the root cause of the failure, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- id: 1, check: Verify the redundant link status between the PE node and the core router using the command 'show ip int brief'.

- id: 2, check: Confirm that the power supply redundancy is intact by checking the status of the power supplies using the command 'show power supply'.

- id: 3, check: Ensure that the remote hands are dispatched to the Koramangala Datacenter to facilitate any necessary physical interventions.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show debug atm interface, Expected Output: ATM interface debugging output, Description: This command enables ATM interface debugging, which provides detailed information about the ATM cells being processed and forwarded by the Line Card. The output will help identify if there are any issues with the ATM protocol or the Line Card's functionality.
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```
Command: show power supply, Expected Output: Power supply status, Description: This command displays the status of the power supplies, including their operational status, voltage, and current. The output will help determine if the power supply redundancy is intact or if there are any issues with the power supplies.

Command: show tech-support, Expected Output: Technical support information, Description: This command generates a technical support file, which contains detailed information about the node's configuration, software, and hardware. The output will help identify any potential issues with the node's software or hardware.
```

Resolution Steps

Step 1: Step: Replace the failed Line Card, Command: no atm interface , Description: This command removes the ATM interface configuration for the failed Line Card. Once removed, the failed Line Card can be removed and replaced with a new one. After replacing the Line Card, reconfigure the ATM interface using the 'atm interface' command.

Step 2: Step: Replace the failed power supply, Command: power supply redundancy, Description: This command enables power supply redundancy, which ensures that the node will switch to the standby power supply in case the primary power supply fails. Once enabled, verify that the power supply redundancy is working correctly using the 'show power supply' command.

Post-Check Verification

```
Command: show ip int brief, Expected Output: Interface status, Description: This command displays the status of all interfaces on the node, including their operational status and IP addresses. The output will help verify that the interfaces are up and functioning correctly.

Command: show atm interface, Expected Output: ATM interface status, Description: This command displays the status of the ATM interfaces, including their operational status and ATM cell counts. The output will help verify that the ATM interfaces are functioning correctly.
```

Roll-back Procedures

- Step: Restore the original Line Card, Command: atm interface , Description: This command reconfigures the ATM interface for the original Line Card. Once restored, verify that the ATM interfaces are functioning correctly using the 'show atm interface' command.

- Step: Disable power supply redundancy, Command: no power supply redundancy, Description: This command disables power supply redundancy, which returns the node to its original configuration. Once disabled, verify that the power supply redundancy is working correctly using the 'show power supply' command.